RomApplication
With
GeoMechanicsApplication

CIMNE

Content

- ROM Idea Overview
- Example Description
- Preliminary Results
- Discussion

ROM Idea Overview

ROM Idea Overview

INPUT

μ

Parameters

Full Order Model (FOM)

KRATOS MULTI-PHYSICS

GeoMechanicsApplication

OUTPUT

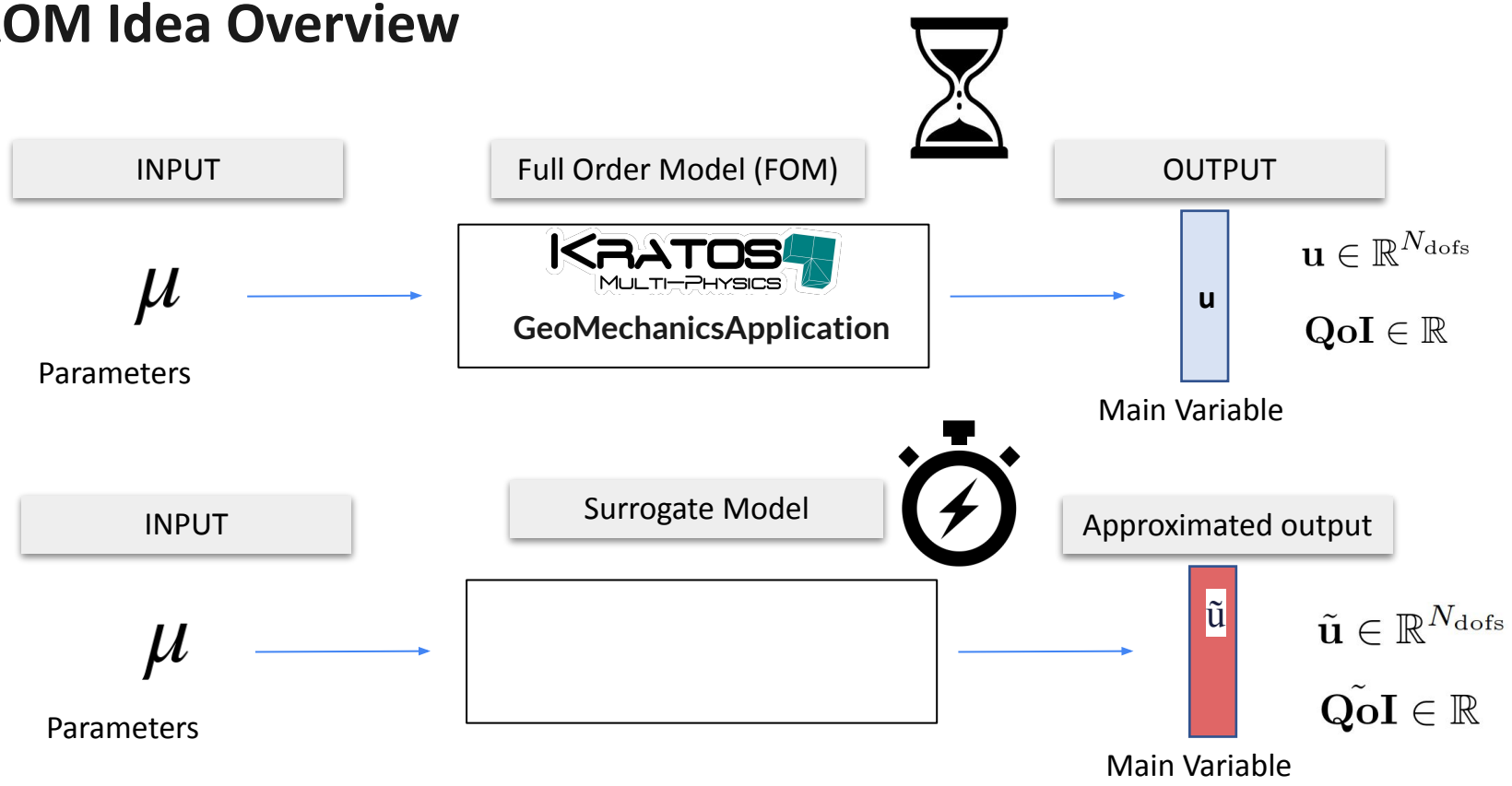
u

Main Variable

$\mathbf{u} \in \mathbb{R}^{N_{\text{dofs}}}$

$\mathbf{QoI} \in \mathbb{R}$

ROM Idea Overview



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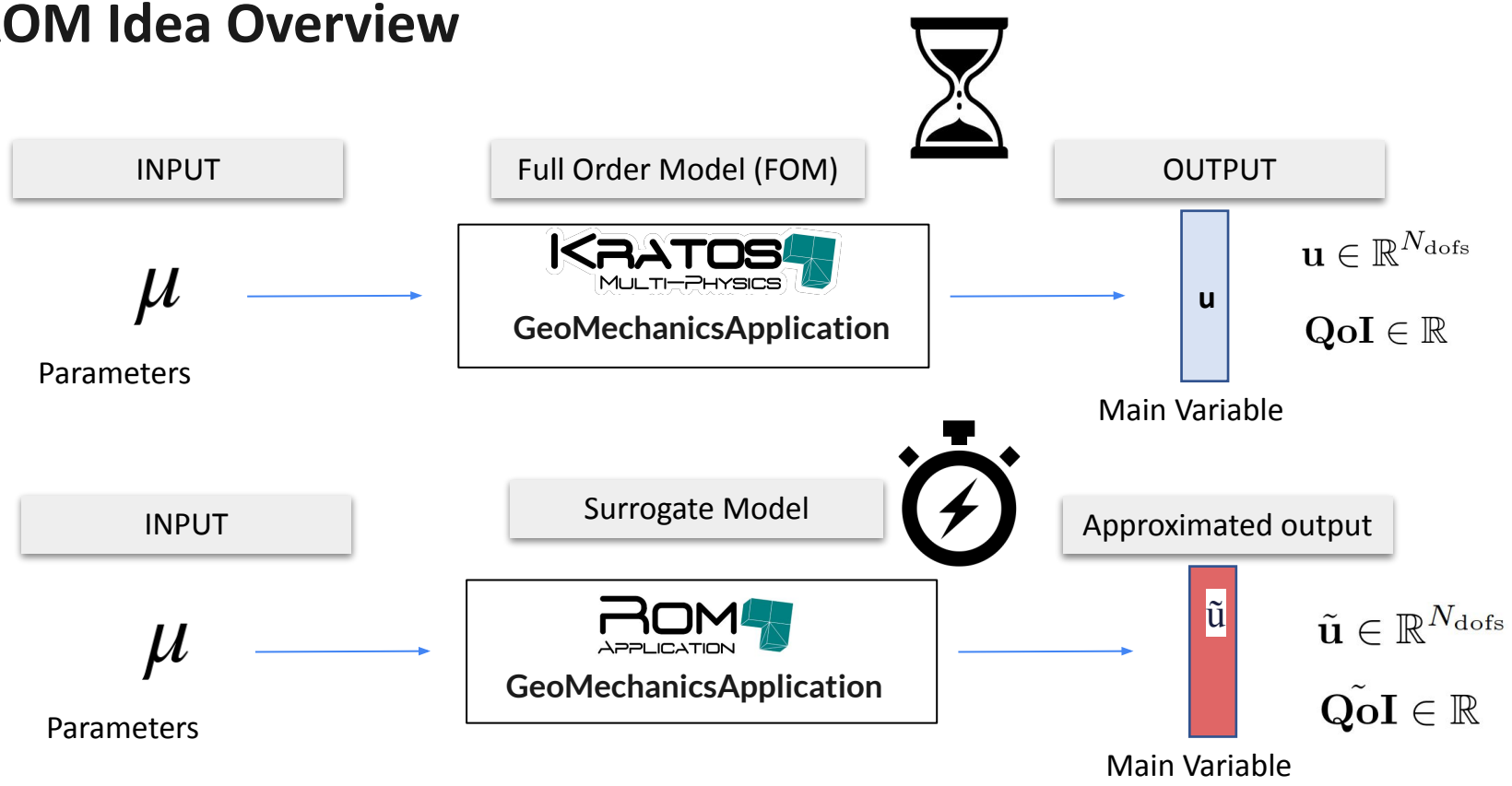

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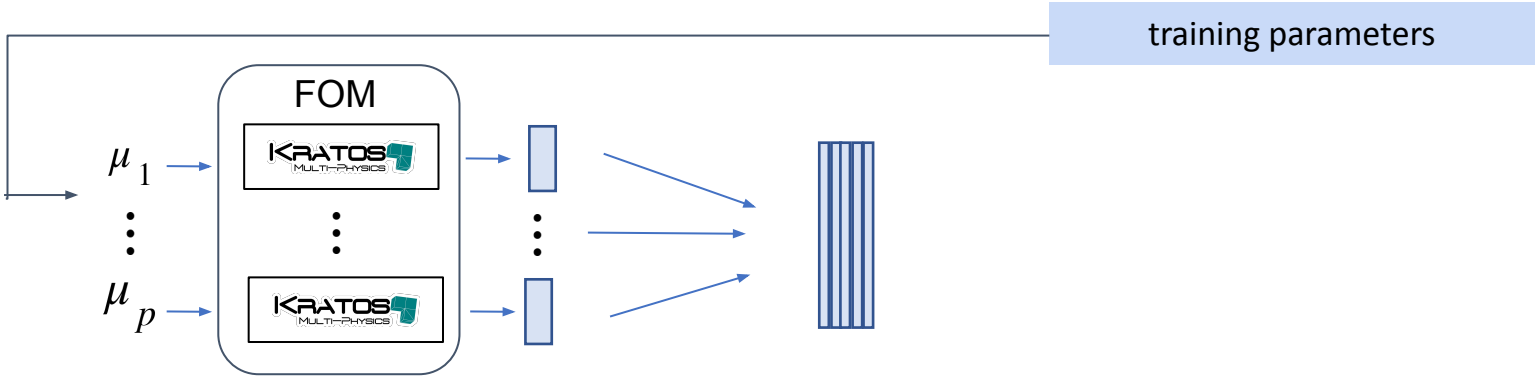

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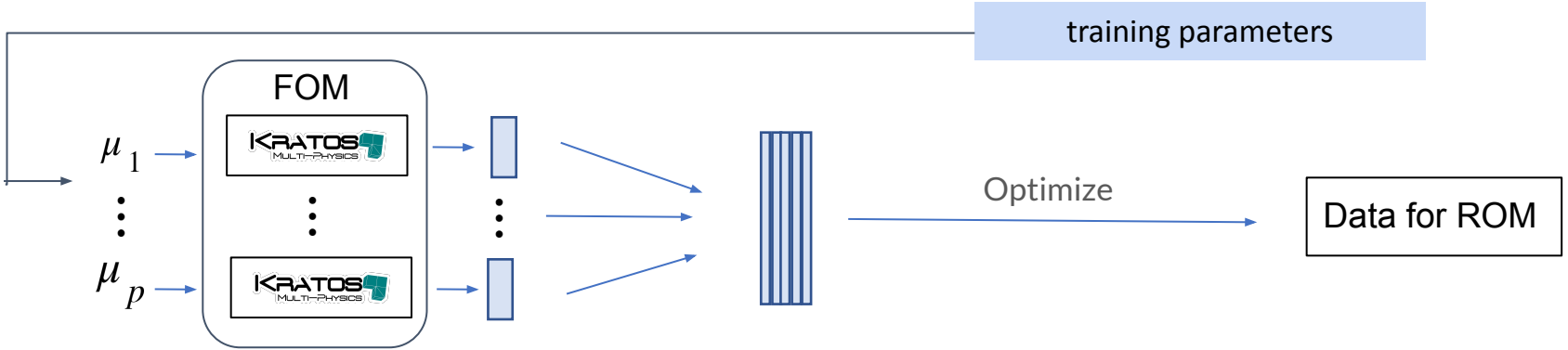

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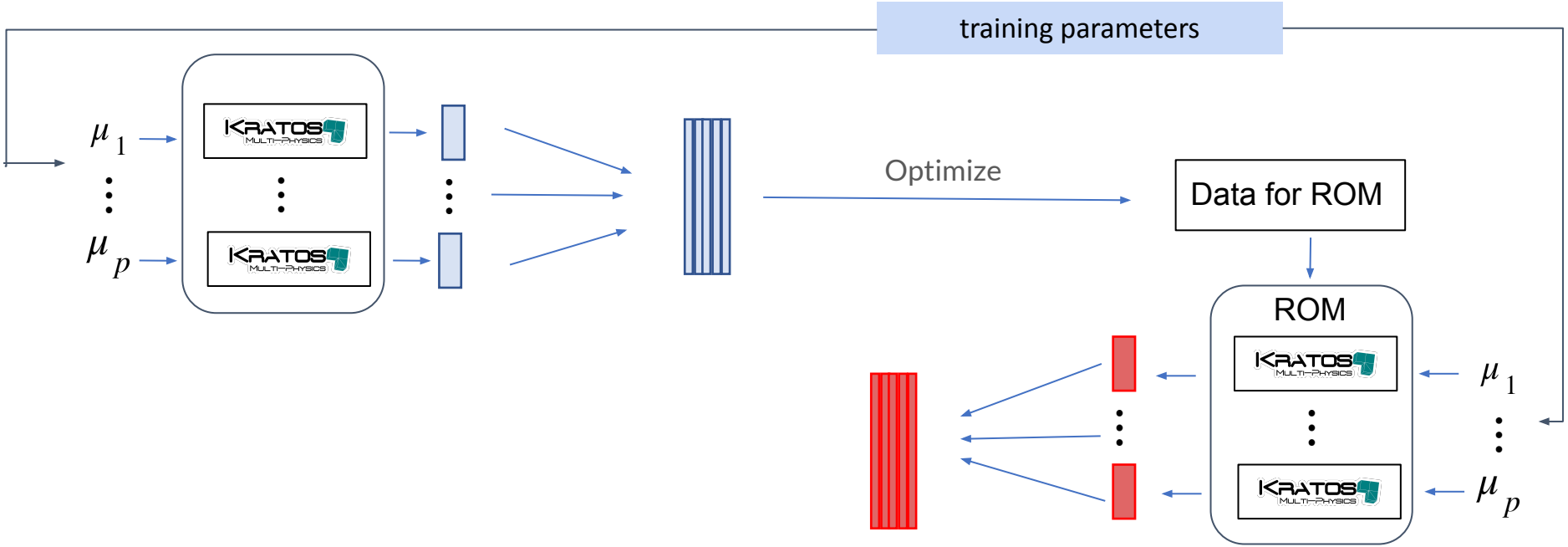

Example Description

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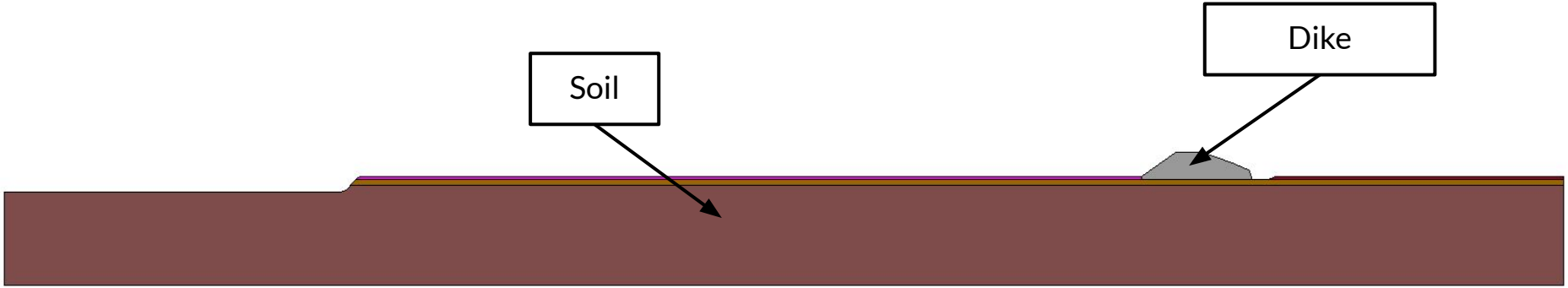
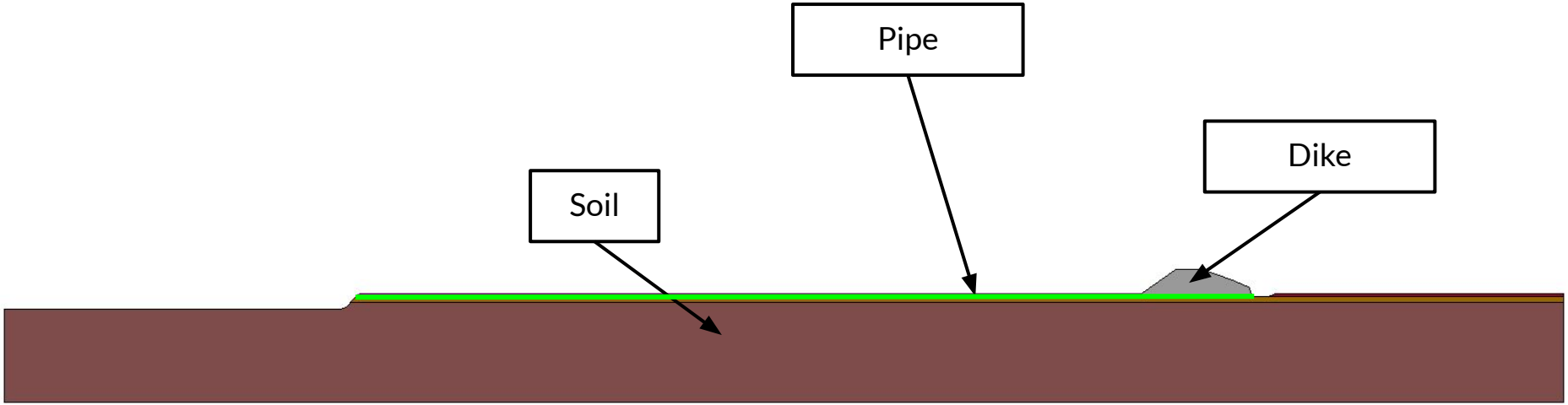
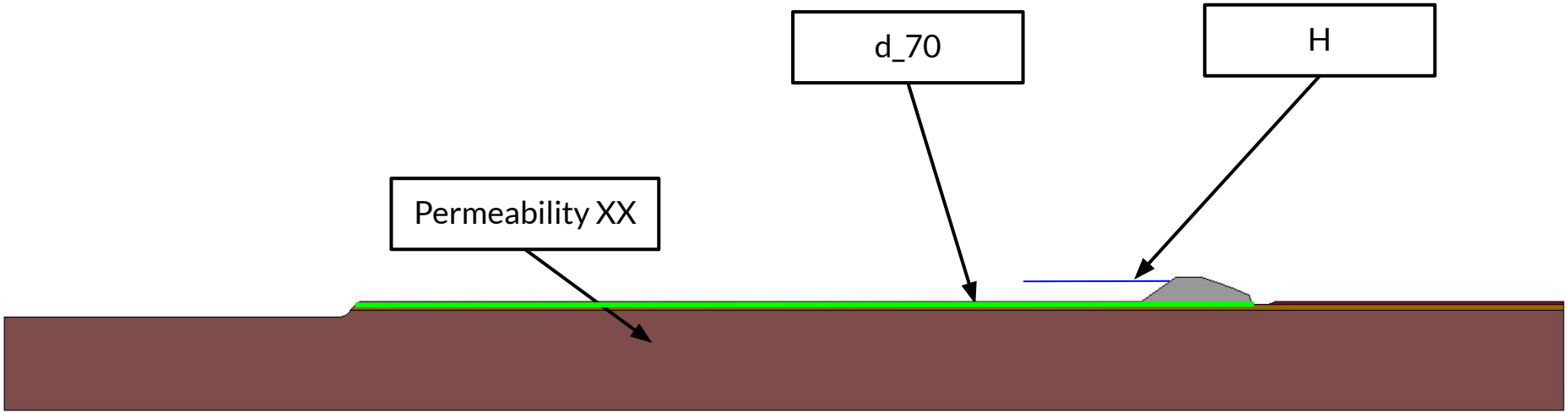


Example Description

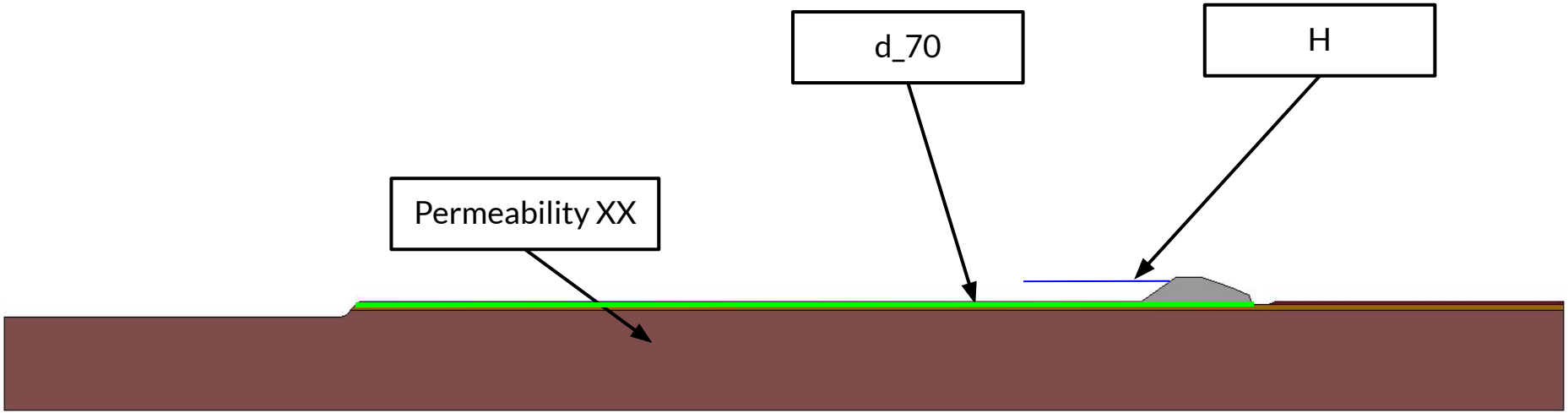


Example Description



$$\boldsymbol{\mu} = \begin{bmatrix} \kappa \\ d_{70} \\ H \end{bmatrix}$$

Example Description



$$\boldsymbol{\mu} = \begin{bmatrix} \kappa \\ d_{70} \\ H \end{bmatrix}$$

$\mathbf{u} \in \mathbb{R}^{N_{\text{dofs}}}$ Water Pressure

$\mathbf{QoI} \in \mathbb{R}$ Pipe length

Preliminary Results

Preliminary Results

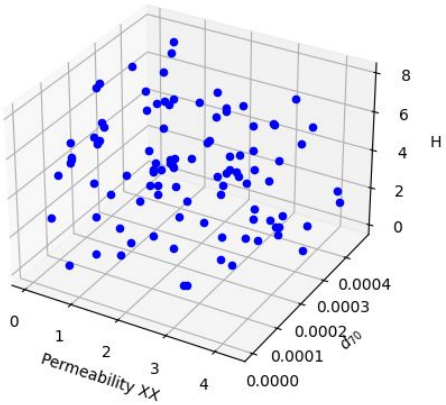
$$\boldsymbol{\mu} = \begin{bmatrix} \kappa \\ d_{70} \\ H \end{bmatrix}$$

$$\kappa_{\mathrm{xx}} \in [0, 286] \text{ (m/day)}$$

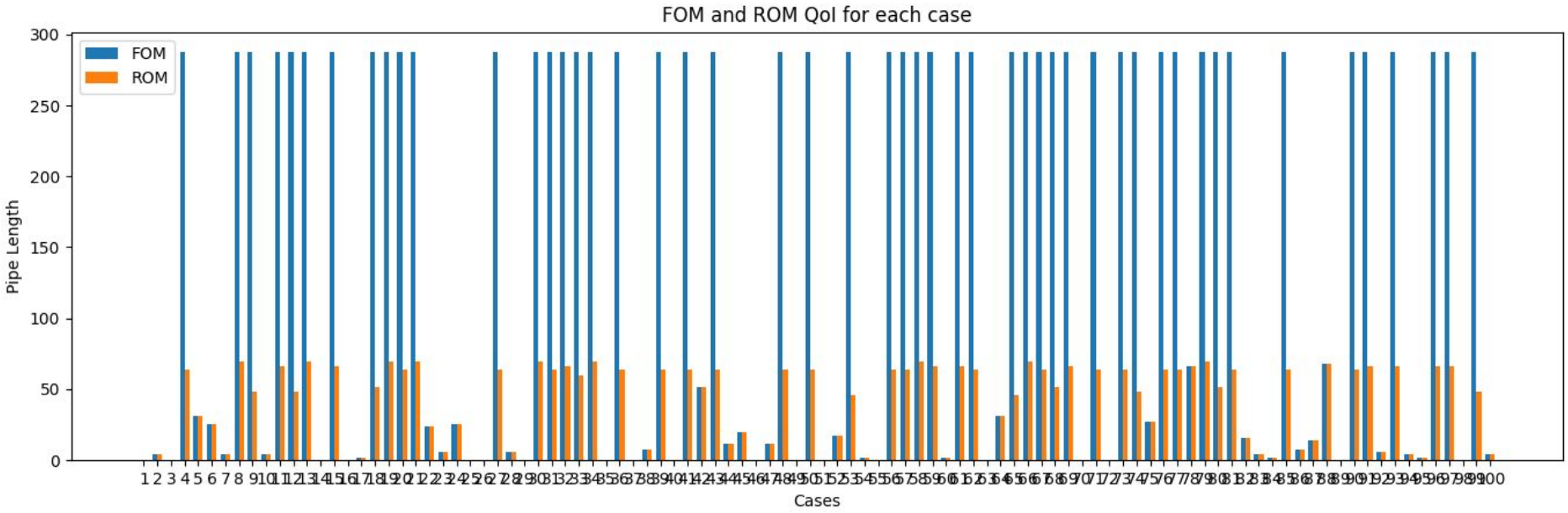
$$d_{70} \in [0, 0.478] \text{ (mm)}$$

$$H \in [0, 8] \text{ (meters)}$$

100 random samples



Preliminary Results



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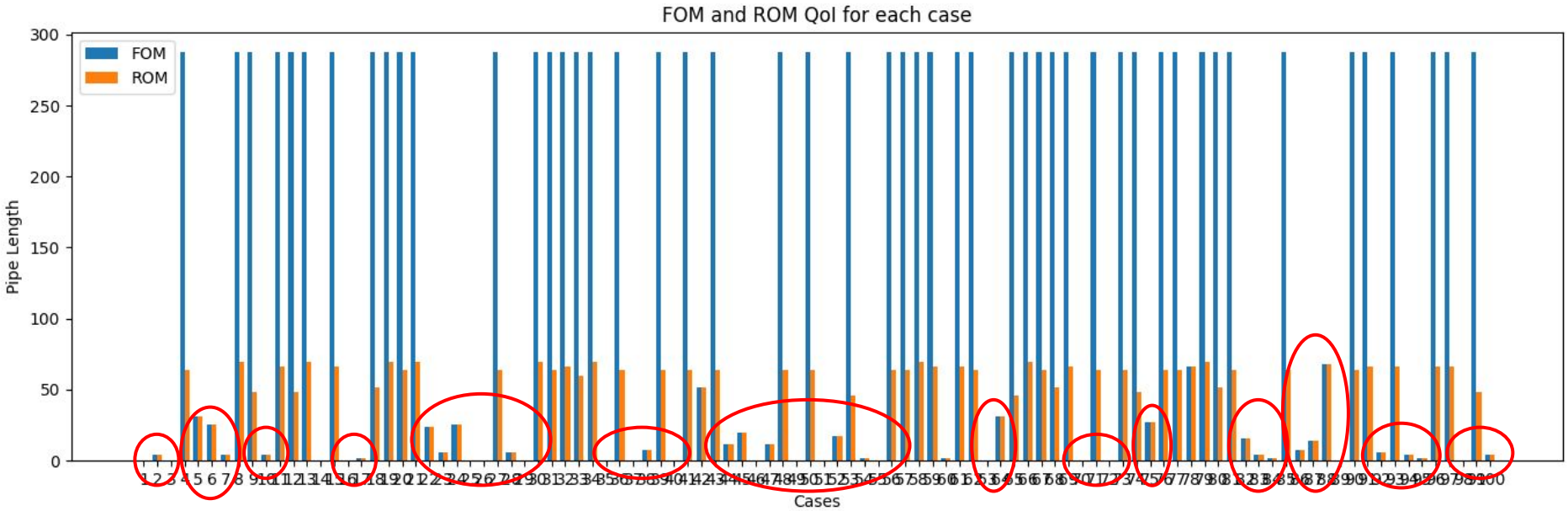
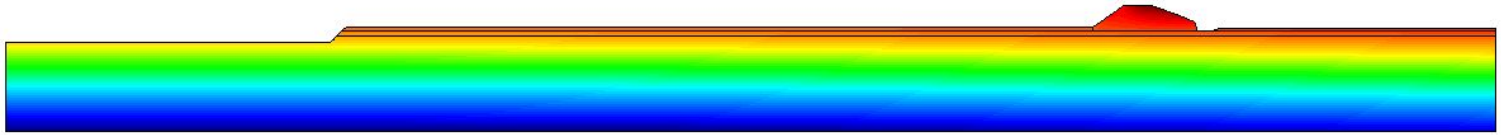


Correctly approximate the cases where the pipe length does not traverse the complete geometry

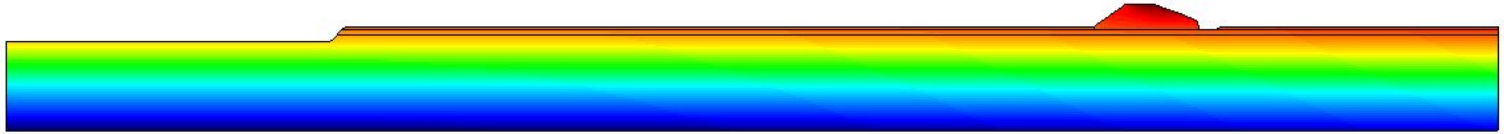
Preliminary Results

κ_{xx} d_{70} H

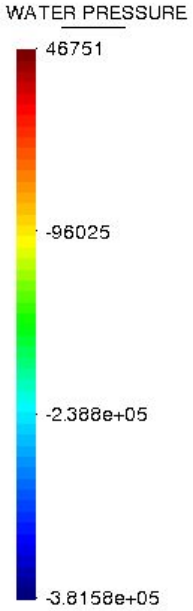
Case 63: [1.551359088177884e-10, 0.00025610029898773457, 4.146637218909894]



FOM Pipe Length: 31.38m



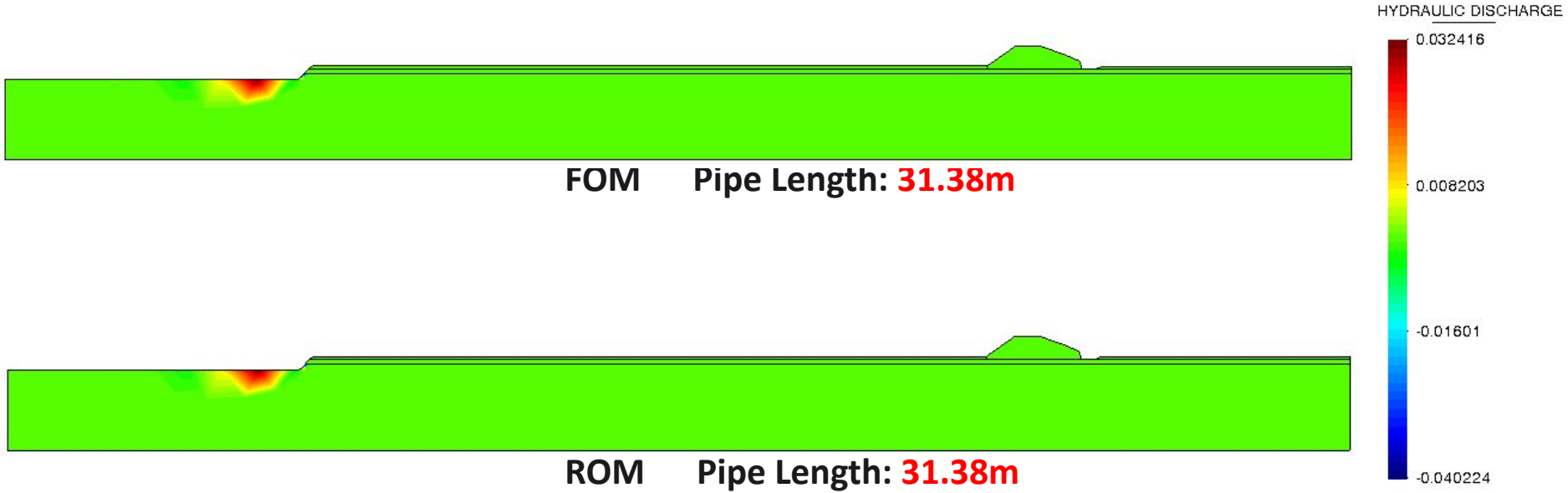
ROM Pipe Length: 31.38m



Preliminary Results

κ_{xx} d_{70} H

Case 63: [1.551359088177884e-10, 0.00025610029898773457, 4.146637218909894]

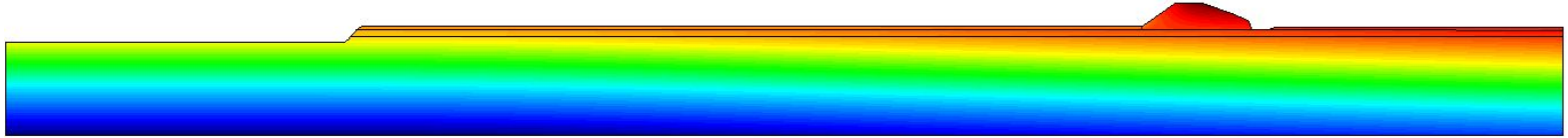


FOM **Pipe Length: 31.38m**

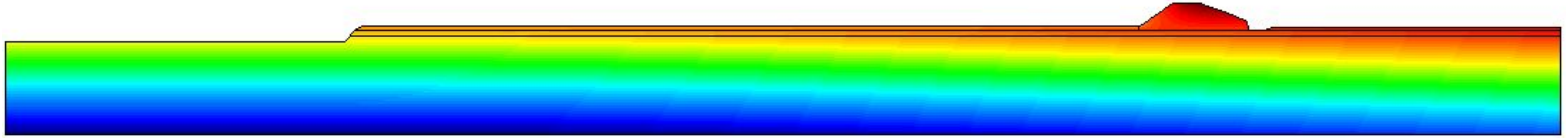
ROM **Pipe Length: 31.38m**

Preliminary Results

	κ_{xx}	d_{70}	H
Case 3:	2.6032953603584655e-10	0.0002430967903767279	7.188433508216634



FOM **Pipe Length: 287.42m**

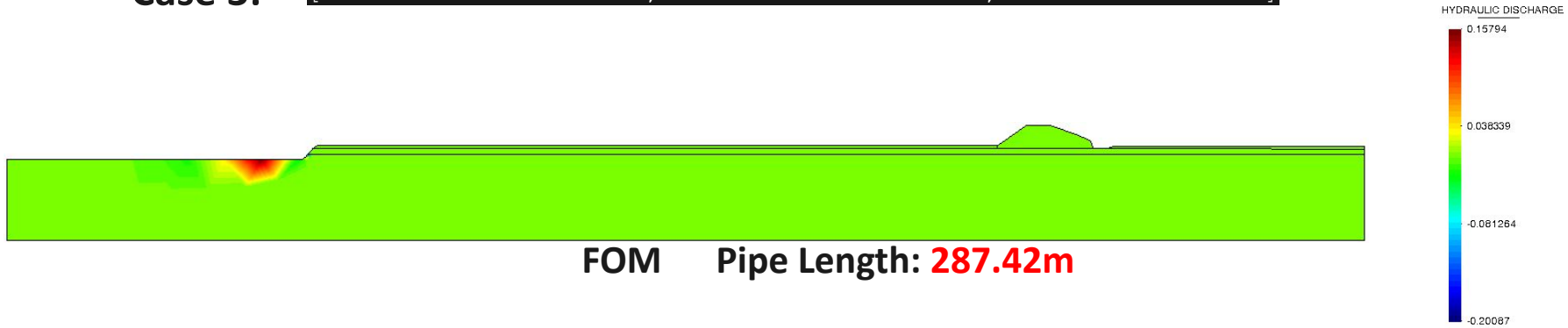


ROM **Pipe Length: 53.97m**

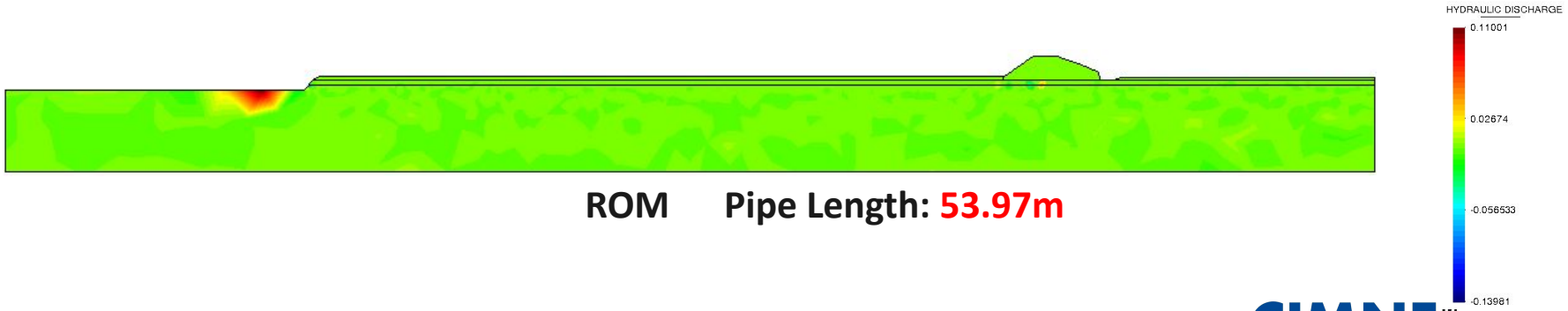
Preliminary Results

κ_{xx} d_{70} H

Case 3: [2.6032953603584655e-10, 0.000243096790376727 9, 7.188433508216634]



FOM **Pipe Length: 287.42m**



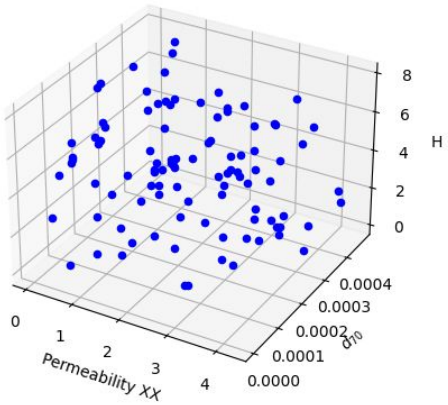
ROM **Pipe Length: 53.97m**

Preliminary Results

$$\boldsymbol{\mu} = \begin{bmatrix} \kappa \\ d_{70} \\ H \end{bmatrix}$$

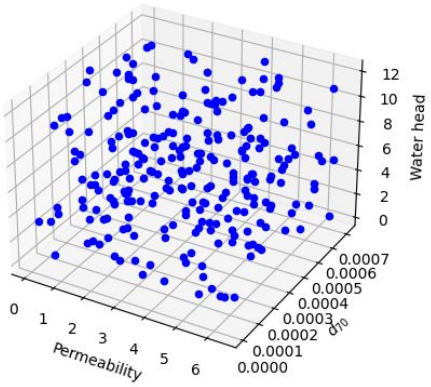
Example 1

100 random samples



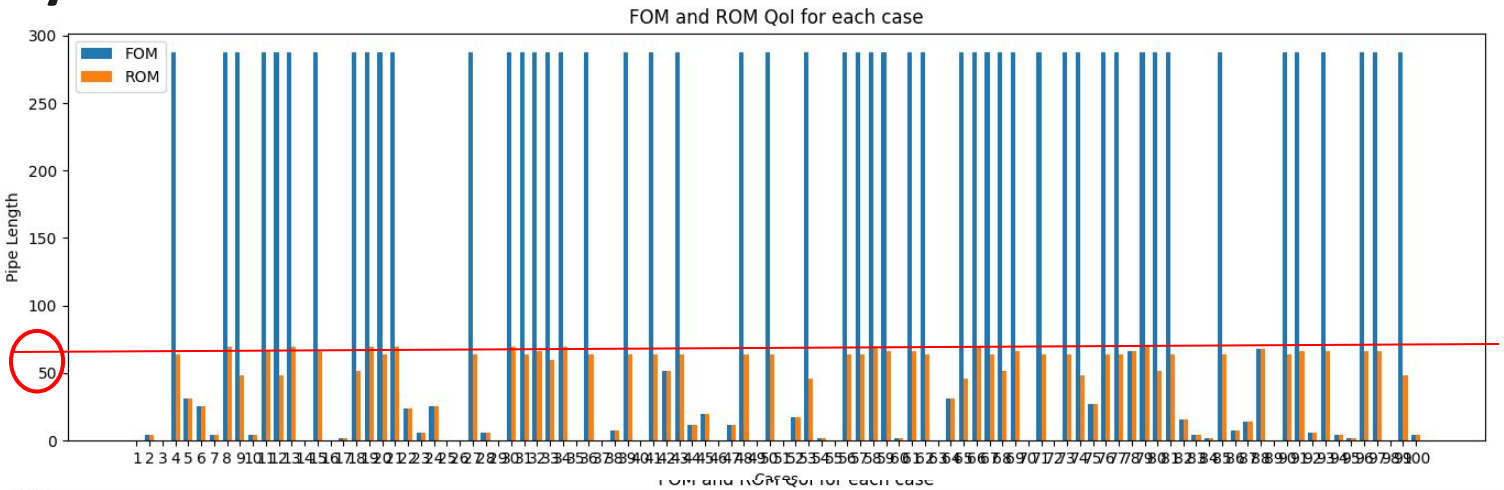
Example 2

250 random samples

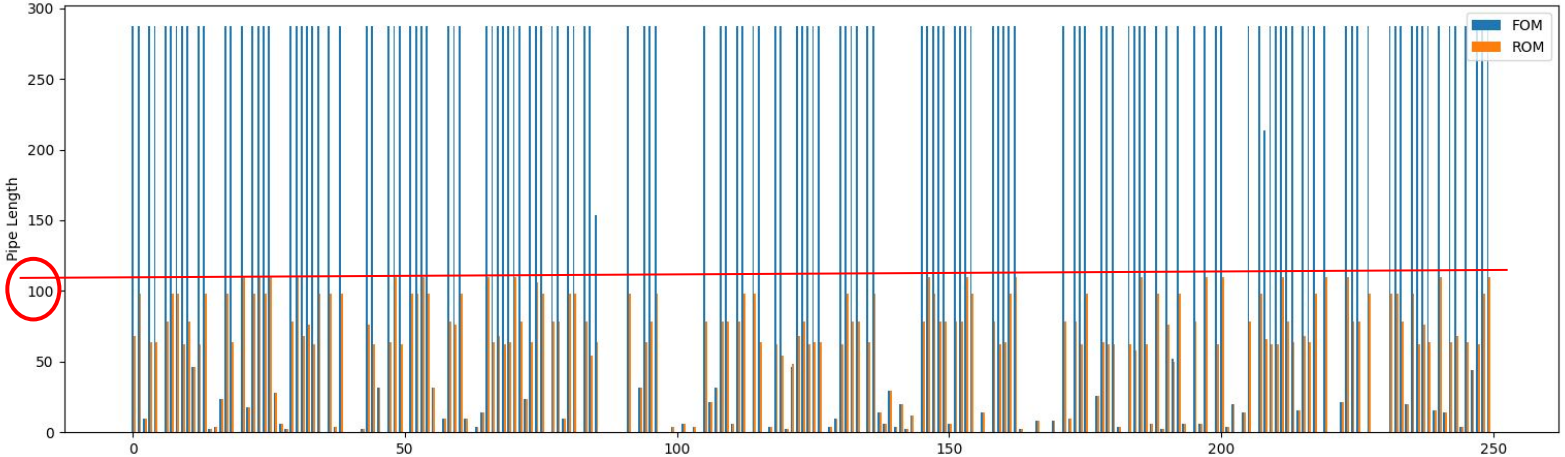


Preliminary Results

Example 1

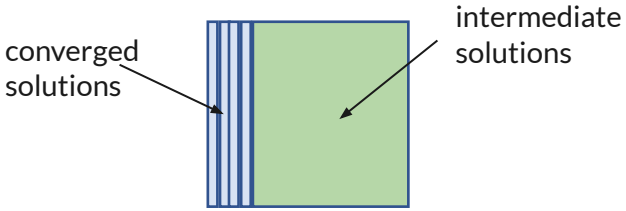


Example 2



Preliminary Results

$$\boldsymbol{\mu} = \begin{bmatrix} \kappa \\ d_{70} \\ H \end{bmatrix}$$



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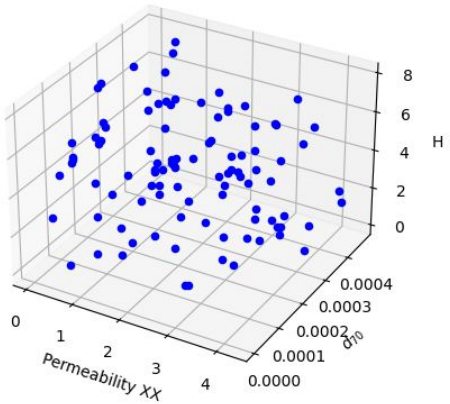
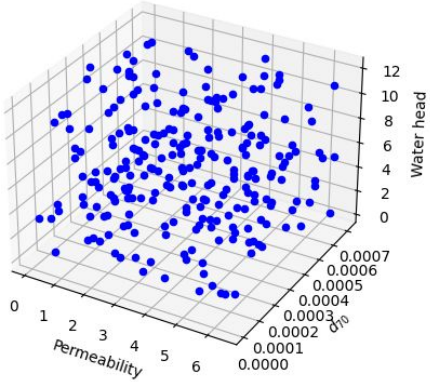
100 random samples



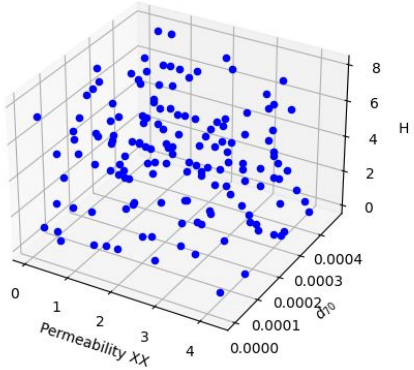
Example 2

250 random samples



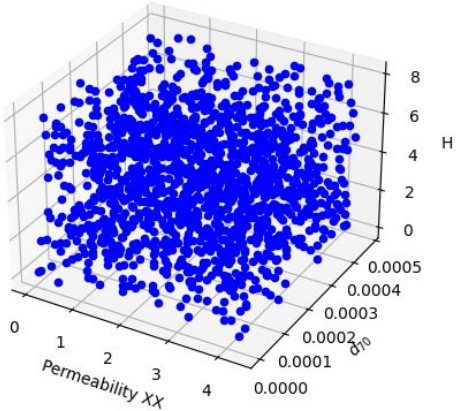
Example 3

150 random samples >>
150 converged sols +
39K intermediate sols



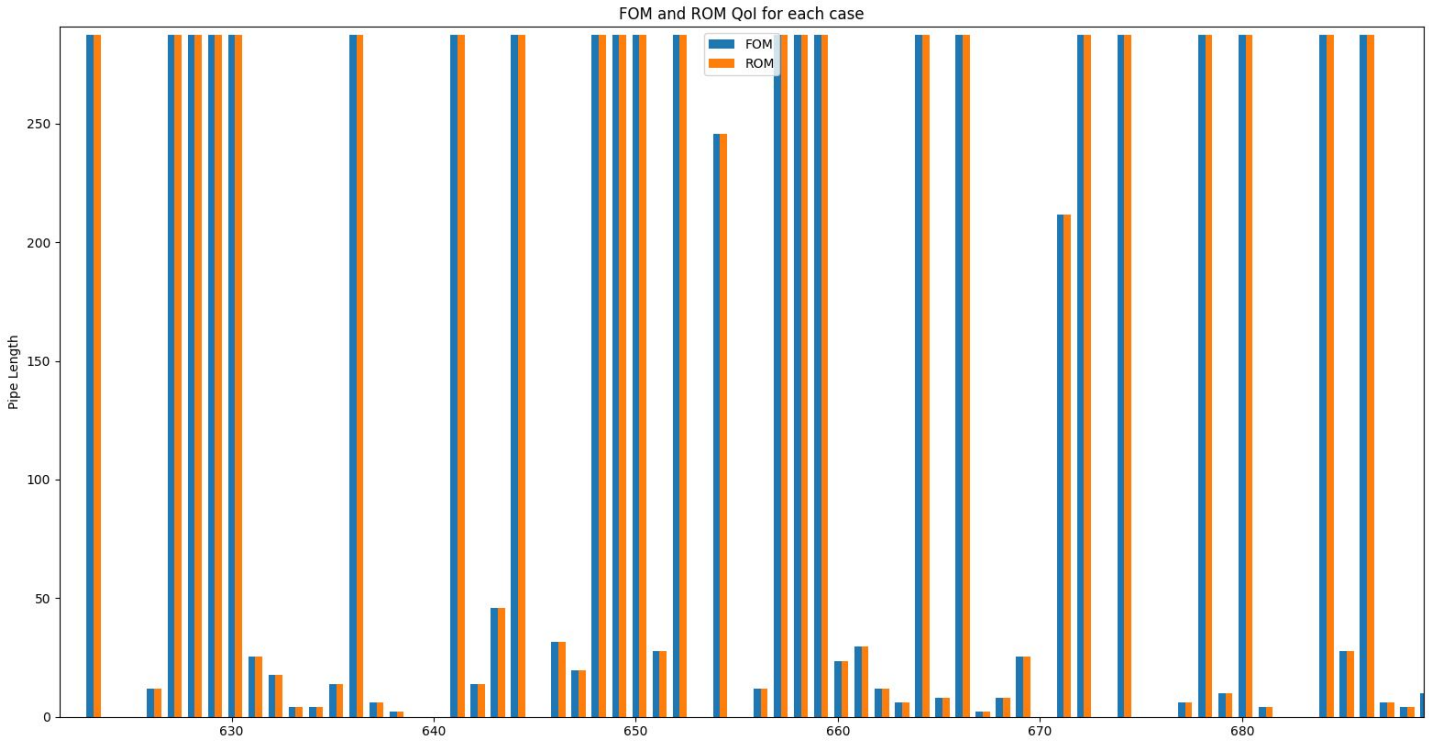
Example 4

1500 random samples +
370K intermediate sols



Preliminary Results

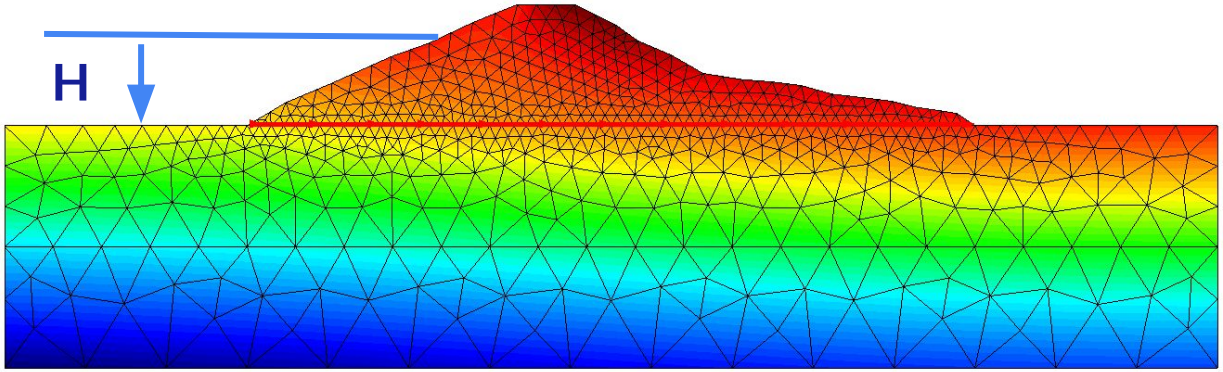
Example 4



Perfect reconstruction of training cases

Critical Head Calculation

- The decision was taken to switch to an example of critical head calculation in order to verify the potential or ROMs without the necessity of intermediate solutions.
- Parameter to vary is the head. Find the maximum value before failure



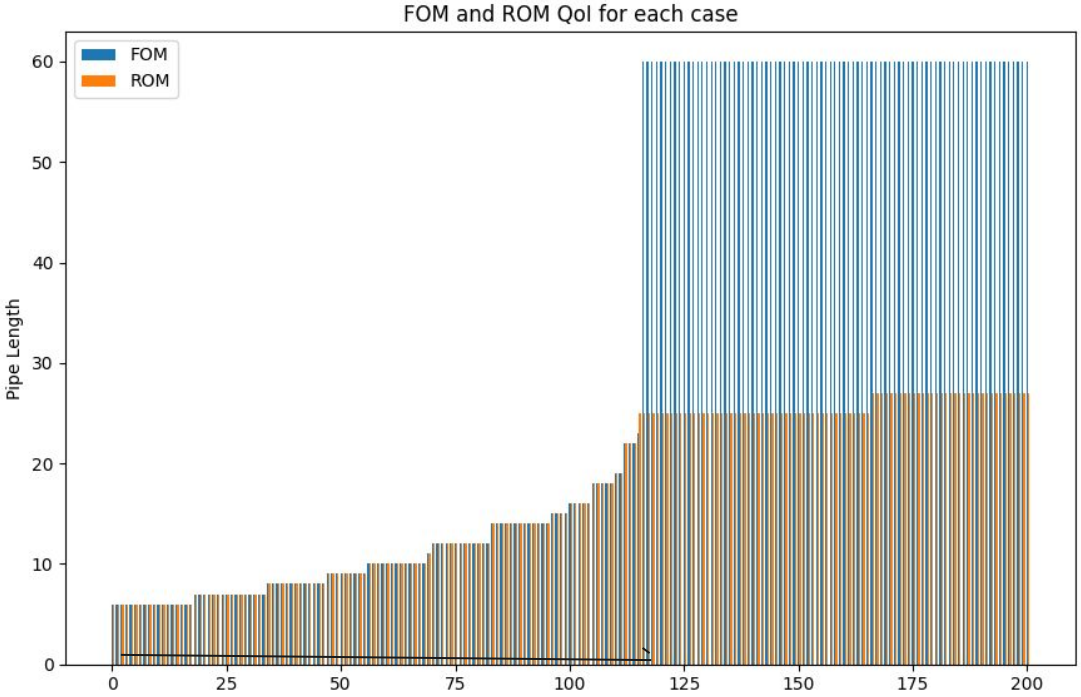
Steps taken

- Train ROM with **converged snapshots** from FOM with different head heights

- Launch critical head (previous slide) script using ROM. Does the ROM predict the critical head of the FOM?

Steps taken:Train

H = [5.01, 5.02, … , 6.98, 6.99, 7.00] Hmin = 5 , Hmax = 7, 200 samples



H[0] **H[199]**

Steps taken:Launch critical head

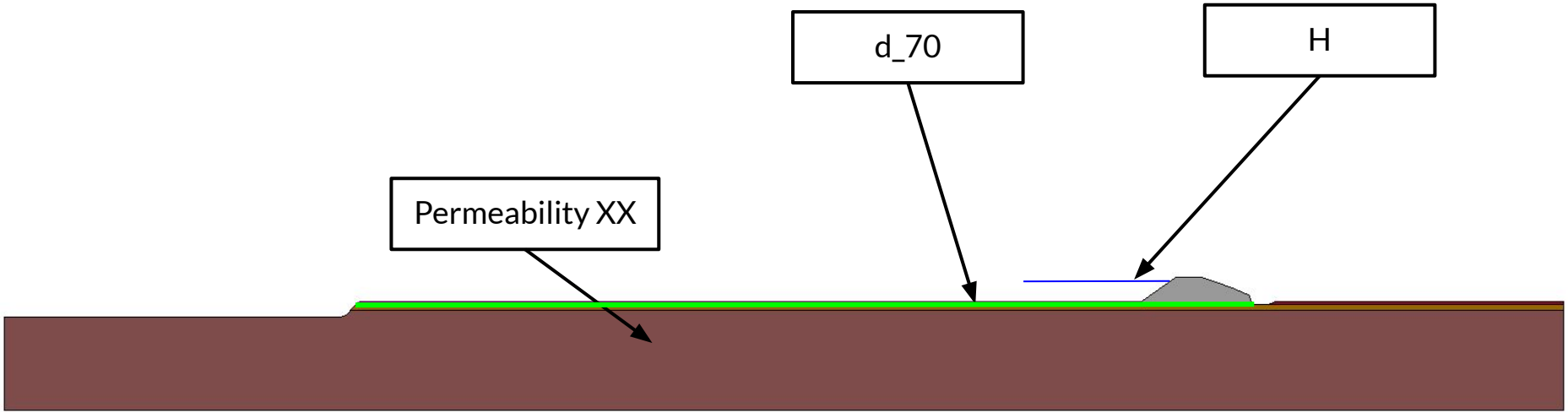
- The ROM does not find the critical head.

Critical head:
FOM: 6.1999 m
ROM: None

- This is because using a simple sampling, the ROM does not develop a complete pipe

Try to set H = 20m —--> ROM Critical head: None

Example Description



$$\boldsymbol{\mu} = \begin{bmatrix} \kappa \\ d_{70} \\ H \end{bmatrix}$$

$\mathbf{u} \in \mathbb{R}^{N_{\text{dofs}}}$ Water Pressure

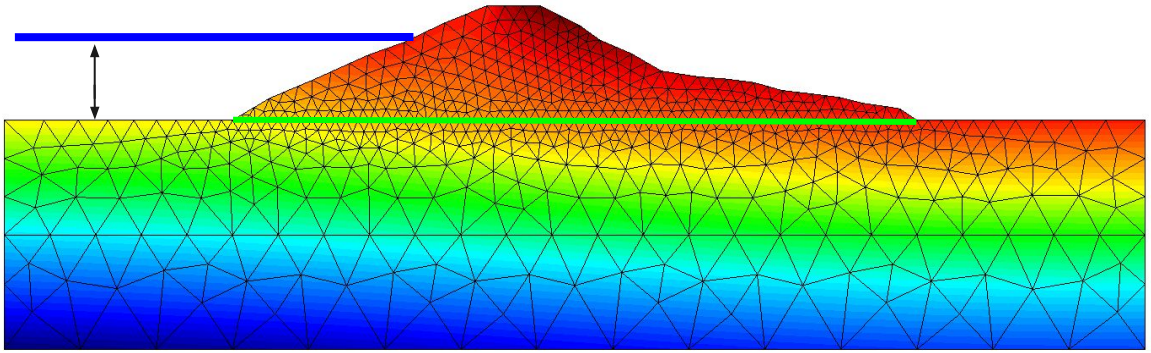
$\mathbf{QoI} \in \mathbb{R}$ Pipe length

Error estimator based on Residuals

Find the critical head H_c. Increase the value of ***H*** until you get a full pipe, then return the previous value of ***H***

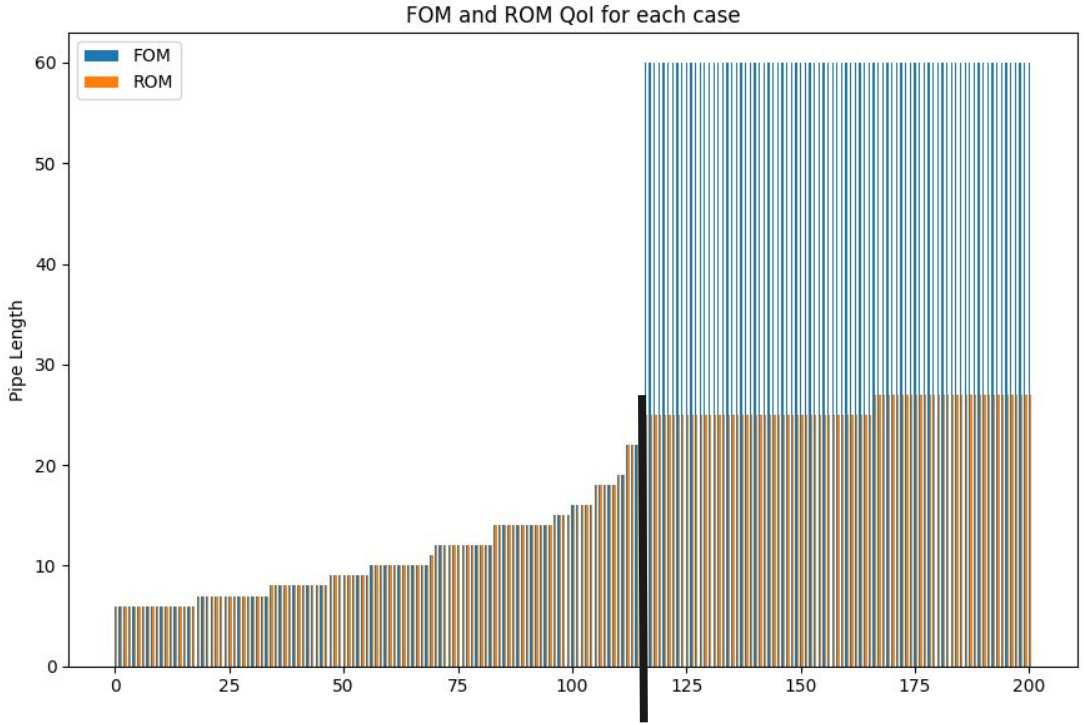
$$\mu = \begin{bmatrix} \texttt{d_70} \\ \texttt{permeability} \end{bmatrix}$$

$$\boxed{\texttt{other params}: H_{\min} \quad H_{\max} \quad \texttt{step}}$$



Error estimator based on Residuals

H = [5.01, 5.02, … , 6.98, 6.99, 7.00] Hmin = 5 , Hmax = 7, step: 0.01, 200 steps

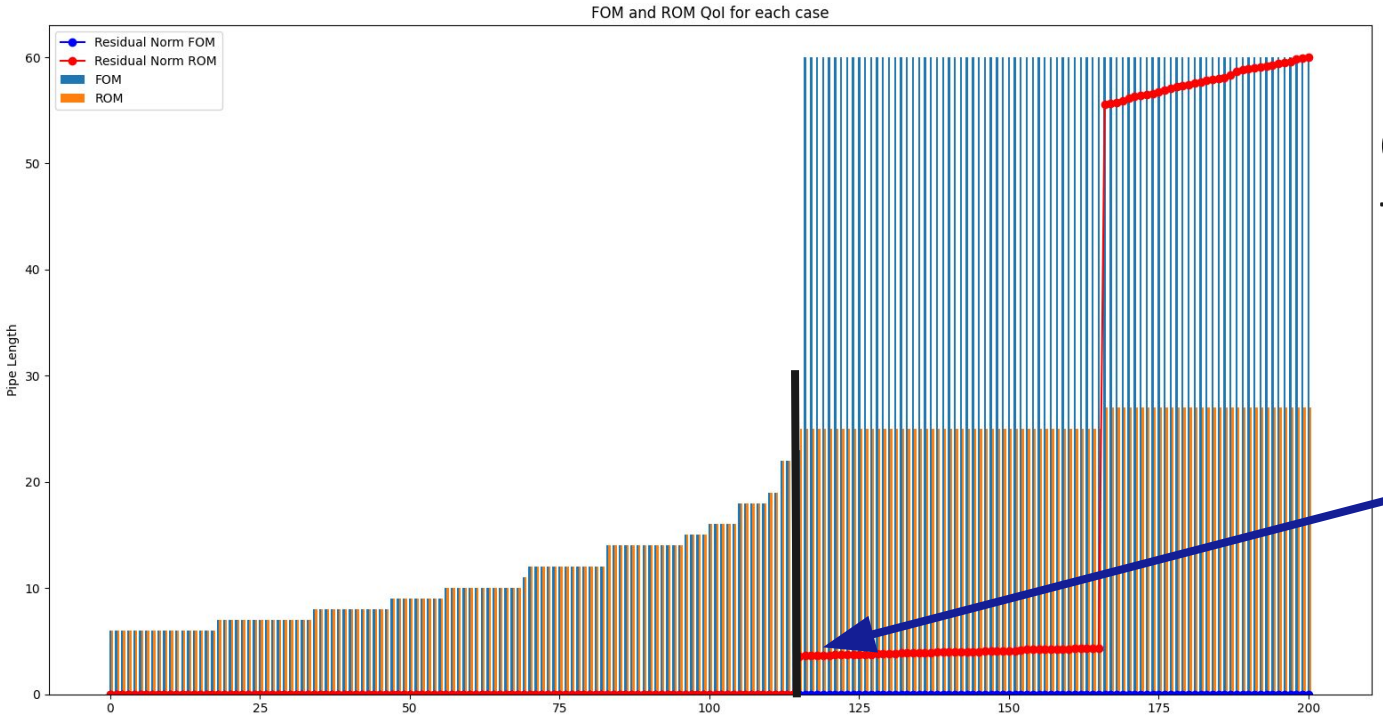


Case for a fixed μ

After H_c , all H produce a full pipe

Error estimator based on Residuals

H = [5.01, 5.02, … , 6.98, 6.99, 7.00] Hmin = 5 , Hmax = 7, step: 0.01, 200 steps



Case for a fixed μ

The residual norm is an indicator of the ROM failing, and for this case, **an indicator of full pipe**

After H_c , all ***H*** produce a full pipe

Error estimator based on Residuals

Find the critical head $\boldsymbol{H}_c$. Increase the value of ***H*** until you get a full pipe, then return the previous value of ***H***

Algorithm 1 FOM critical head

```
Input: μ, H_min, H_max, step
Output: H_c
H ← H_min
full_pipe = Solve_FOM(H, μ)
while H < H_max do
    if full_pipe then
        return H - step
    end if
    H ← H + step
    full_pipe = Solve_FOM(H, μ)
end while
```

Error estimator based on Residuals

Results are sensible to value of threshold tolerance.

Algorithm 1 FOM critical head

```
Input: μ, H_min, H_max, step
Output: H_c
H ← H_min
full_pipe = Solve_FOM(H, μ)
while H < H_max do
    if full_pipe then
        return H - step
    end if
    H ← H + step
    full_pipe = Solve_FOM(H, μ)
end while
```

Algorithm 2 ROM critical head

```
Input: μ, H_min, H_max, step
Output: H_c
H ← H_min
residual_norm = Solve_ROM(H, μ)
while H < H_max do
    if residual_norm > tol then
        return H - step
    end if
    H ← H + step
    residual_norm = Solve_ROM(H, μ)
end while
```